



1.

$$(a) T_1: B'C$$

$$T_2: A'B$$

$$T_3: A + T_1 = A + B'C$$

$$T_4: T_2 \oplus D = A'B \oplus D = A'BD' + (A'B)'D = AD + B'D + A'BD'$$

$$F_1 = T_3 + T_4$$

$$= A + B'C + AD + B'D + A'BD' = A + AD + A'BD' + B'C + B'D$$

$$= A + A'BD' + B'C + B'D = A + BD' + B'C + B'D$$

$$F_2 = T_2 + D' = A'B + D'$$

(b)

A	B	C	D	$T_1(B'C)$	$T_2(A'B)$	$T_3(A+T_1)$	$T_4(T_2 \oplus D)$	F_1	F_2
0	0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	1	1	0
0	0	1	0	1	0	1	0	1	1
0	0	1	1	1	0	1	1	1	0
0	1	0	0	0	1	0	1	1	1
0	1	0	1	0	1	0	0	0	1
0	1	1	0	0	1	0	1	1	1
0	1	1	1	0	1	0	0	0	1
1	0	0	0	0	0	1	0	1	1
1	0	0	1	0	0	1	1	1	0
1	0	1	0	1	0	1	0	1	1
1	0	1	1	1	0	1	1	1	0
1	1	0	0	0	0	1	0	1	1
1	1	0	1	0	0	1	1	1	0
1	1	1	0	0	0	1	0	1	1
1	1	1	1	0	0	1	1	1	0

(C)

F_1

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	0	0	1
11	1	1	1	1
10	1	1	1	1

A

$B'D$

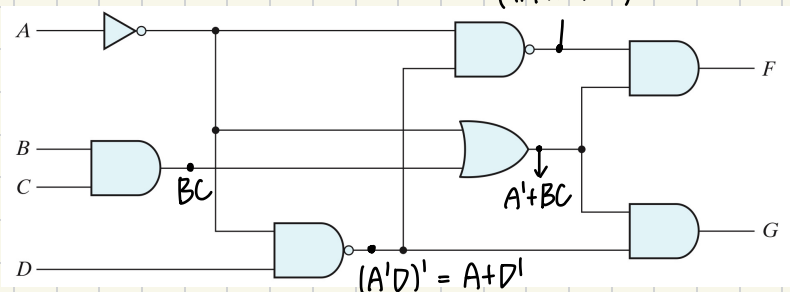
$$A + B'D + BD' + CD'$$

F_2

CD' AB \ CD	00	01	11	10
00	1	0	0	1
01	1	1	1	1
11	1	0	0	1
10	1	0	0	1

$$D' + A'B$$

$$((A+D')A')' = A+D$$



$$F = (A+D)(A'+BC) = ABC + A'D + BCD$$

$$G = (A'+BC)(A+D') = A'D' + ABC + BCD'$$

$$F = ABC + A'D + BCD$$

111X 0xx1 x111

$$G = A'D' + ABC + BCD'$$

0xx0 111X x110

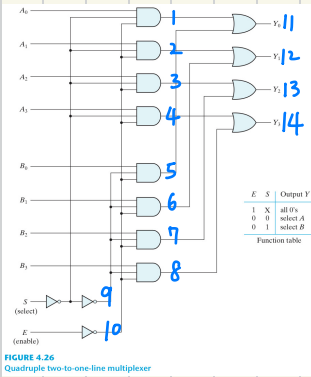
AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11				1
10				1

$$F = A'C + ABC$$

AB \ CD	00	01	11	10
00	1			1
01	1			
11			1	1
10				

$$F = A'D' + ABC$$

3,



$$Y_i = (A_i S' + B_i S) E', i = 0, 1, 2, 3$$

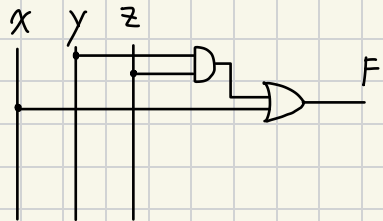
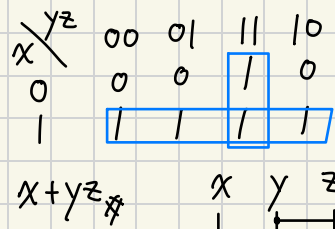
$$2^{10} = 1024$$

1024 rows & 14 columns #

4

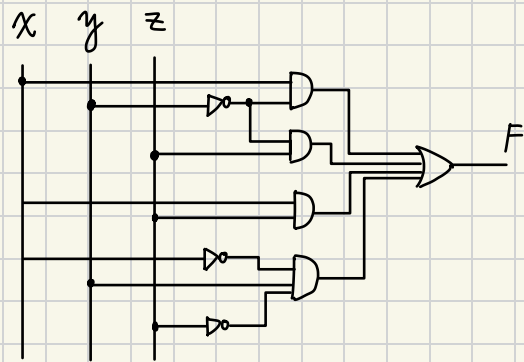
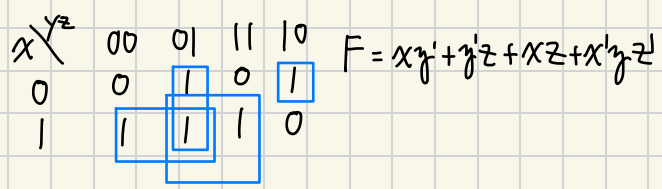
(a)

x	y	z	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



(b)

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



7.

Table 1.6
Gray Code

Gray Code	wxyz	Decimal Equivalent
0000	0000	0
0001	0001	1
0011	0010	2
0010	0011	3
0110	0100	4
0111	0101	5
0101	0110	6
0100	0111	7
1100	1000	8
1101	1001	9
1111	1010	10
1110	1011	11
1010	1100	12
1011	1101	13
1001	1110	14
1000	1111	15

W:

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

A

X:

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	1	1	1

$$AB' + A'B = A \oplus B$$

Y:

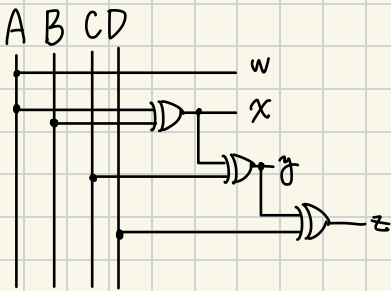
AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	0	0	1	1
10	1	1	0	0

Z:

AB \ CD	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

$$A \oplus B \oplus C \oplus D$$

$$\begin{aligned}
 &A'B'C + A'BC' + ABC + ABC' \\
 &= A'(B'C + BC') + A(BC + B'C') \\
 &= A'(B \oplus C) + A(B \oplus C)' \\
 &= A \oplus B \oplus C
 \end{aligned}$$



10.

2's complement

A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	0	1	1	0
1	0	1	1	0	1	0	1
1	1	0	0	0	1	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	0	1	0
1	1	1	1	0	0	0	1

W:

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	0	0	0	0
10	1	0	0	0

$$\begin{aligned}
 & A'B + A'D + A'C + AB'C'D' \\
 &= A'(B+C+D) + AB'C'D' \\
 &= A \oplus (B+C+D)
 \end{aligned}$$

Y:

AB \ CD	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	0	1	0	1
10	0	1	0	1

$$\begin{aligned}
 & C'D + CD' \\
 &= C \oplus D
 \end{aligned}$$

X:

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	1	0	0	0
10	0	1	1	1

$$\begin{aligned}
 & BC'D' + B'D + B'C \\
 &= BC'D' + B'(C+D) \\
 &= B \oplus (C+D)
 \end{aligned}$$

Z:

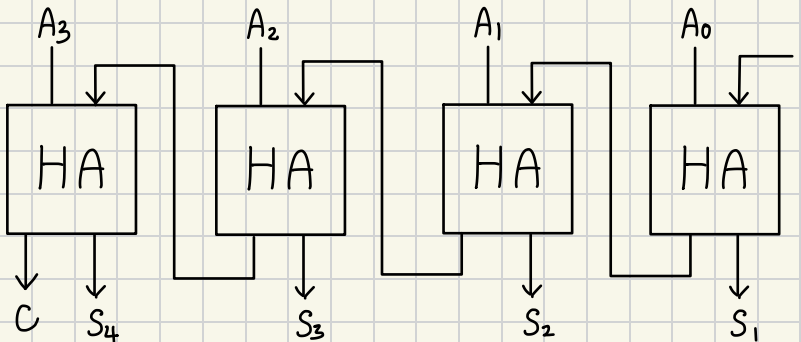
AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

D

five Bit 2's complement : $E \oplus (A+B+C+D)$

11.

(a)



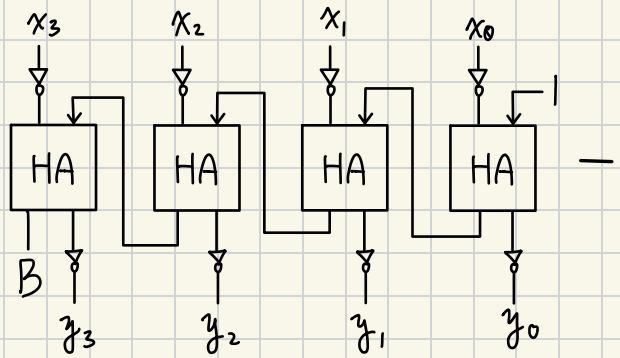
(b)

$$y_0 = x_0 \oplus 1 = x_0'$$

$$y_i = x_i \oplus B_i, \quad i = 1, 2, 3, \dots$$

$$B_0 = 1, \quad B_{i+1} = x_i' \oplus B_i$$

(減1)



12

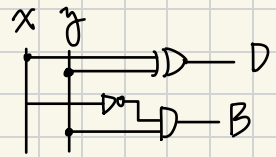
(a)

x	y	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

D: x \ y	0	1
0	0	1
1	1	0

$$x'y + xy' = x \oplus y$$

B: x \ y	0	1
0	0	1
1	0	0

$$B = x'y$$


(b)

x	y	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

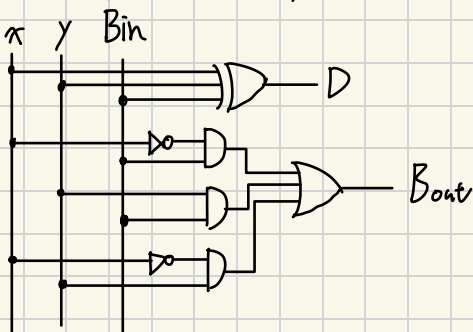
D: x \ y Bin	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$xy' + x'y$$

$$xy'Bin' + x'y'Bin + xyBin + x'yBin' = Bin'(x \oplus y) + Bin(x \oplus y)' = x \oplus y \oplus Bin$$

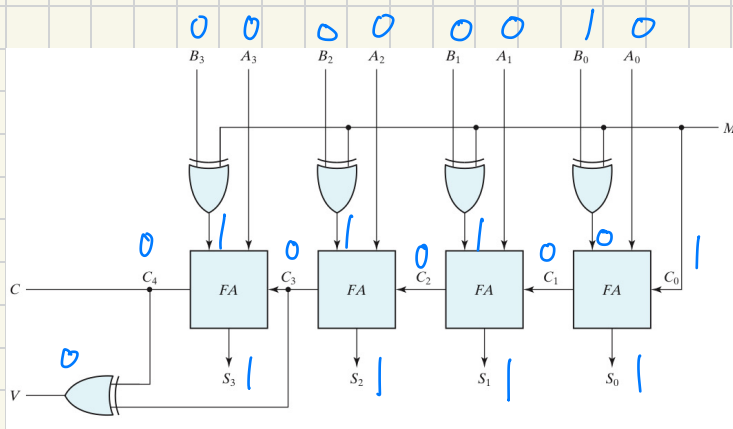
Bout: x \ y Bin	00	01	11	10
0	0	1	1	1
1	0	0	1	0

$$x'Bin + yBin + x'y$$



13.

$M=0$, 加法
 $M=1$, 减法

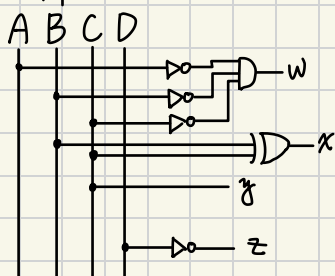


A	B	Sum	C	V (第三 & 第四進位XOR)
0111	+ 0110	1101	0	1
1000	+ 1001	0001	1	1
1100	- 1000	0100	1	0
0101	- 1010	1011	0	1
0000	- 0001	1111	0	0

18.

(a)

A	B	C	D	w	x	y	z
0	0	0	0	0	1	0	0
1	0	0	0	1	1	0	0
2	0	0	1	0	0	1	1
3	0	0	1	1	0	1	0
4	0	1	0	0	0	1	0
5	0	1	0	1	0	1	0
6	0	1	1	0	0	0	1
7	0	1	1	1	0	0	1
8	1	0	0	0	0	0	0
9	1	0	0	1	0	0	0



w:

AB \ CD	00	01	11	10
00	1	1	0	0
01	0	0	0	0
11	x	x	x	x
10	0	0	x	x

$A'B'C'$

x:

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	x	x	x	x
10	0	0	x	x

$BC' + B'C$
 $= B \oplus C$

y:

AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	x	x	x	x
10	0	0	x	x

C

z:

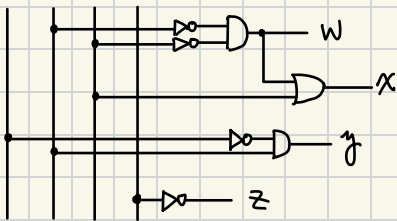
AB \ CD	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	x	x	x	x
10	1	0	x	x

D'

(b)

DEC	A	B	C	D	W	x	y	z
0	0	0	0	0	1	1	0	1
1	0	0	0	1	1	1	0	0
2	0	0	1	1	0	1	0	0
3	0	0	1	0	0	1	0	1
4	0	1	1	0	0	1	1	1
5	0	1	1	1	0	1	1	0
6	0	1	0	1	0	0	1	0
7	0	1	0	0	0	0	1	1
8	1	1	0	0	0	0	0	1
9	1	1	0	1	0	0	0	0

A B C D



w:

AB \ CD	00	01	11	10
00	1	1	0	0
01	0	0	0	0
11	0	0	x	x
10	x	x	x	x

 $B'C'$

y:

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	x	x
10	x	x	x	x

 $A'B$

x:

AB \ CD	00	01	11	10
00	1	1	1	1
01	0	0	1	1
11	0	0	x	x
10	x	x	x	x

 $C+B'C'$

z:

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	x	x
10	x	x	x	x

 D'

27

F_1

A \ BC	00	01	11	10
0	0	1	0	0
1	1	0	0	1

$$\{A'BC + AB'C' + ABC'\}'$$

$$= \{(A'BC)'(AB'C')'(ABC')'\}'$$

011 100 110

F_2

A \ BC	00	01	11	10
0	0	0	1	0
1	0	1	0	0

$$\{ABC + AB'C'\}'$$

$$= \{(A'BC)'(AB'C')'\}'$$

011 101

F_3

A \ BC	00	01	11	10
0	0	0	0	1
1	1	0	1	1

$$\{A'BC' + AB'C' + ABC + ABC'\}'$$

$$= \{(A'BC')'(AB'C')'(ABC)'(ABC')'\}'$$

010 100 111 110

